

Candidate Number:

BAULKHAM HILLS HIGH SCHOOL

Higher School Certificate

2010

Trial Examination

Mathematics

Extension I

General Instructions

- Exam time – 2 hours
- Reading time – 5 minutes
- Start each question on a new page
- All necessary working should be shown
- Write your student number at the top of each page of your answer booklet
- Board approved calculators may be used
- Write, using black or blue pen

Total Marks: 84

Attempt ALL questions

Question 1 (12 marks)**Marks**

- a) Evaluate $\int_0^{\frac{\pi}{2}} \cos^2 x \, dx$ 2
- b) Solve the equation $\cos 2\theta = \sin \theta$ for $0 \leq \theta \leq 2\pi$ 3
- c) Solve the inequality $\frac{3x+4}{x-5} \geq 2$ 2
- d) By using the substitution $w = t^2 - 2$, evaluate $\int_{-1}^{14} \frac{w \, dw}{\sqrt{w+2}}$ 3
- e) A group of six goats is to be chosen from 10 goats.
In how many ways can the group be chosen if: 2
- i) 2 particular goats are included in the group
 - ii) 1 particular goat is excluded from the group

Question 2 (12 marks) - Start a new page

- a) A (4,10), B (-3,1), C (5,7) are the vertices of triangle ABC and E is the midpoint of the side BC. 3
- Find the value of $\tan \theta$ where $\theta = \angle AEC$
- b) If α , β and γ are the roots of the equation $2x^3 + 5x - 3 = 0$ find the values of
- i) $\alpha + \beta + \gamma$ 1
 - ii) $\alpha^2 + \beta^2 + \gamma^2$ 1
 - iii) $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}$ 1
- c) P(x) is a monic polynomial of degree 3. P(x) has the quadratic factor $x^2 - 1$ and when P(x) is divided by $x - 2$ the remainder is -9. 2
- Form an equation for P(x) and hence solve $P(x) = 0$
- d) For the expansion $\left(x^2 + \frac{4}{x}\right)^{30}$ find which term
- i) is independent of x 2
 - ii) has the greatest coefficient 2

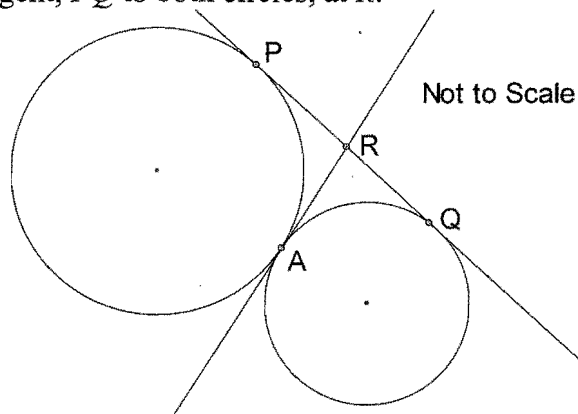
Question 3 (12 marks) - Start a new page**Marks**

- a) Find $\frac{d}{dx}(\tan^{-1} x)^2$ and hence evaluate

3

$$\int_{-1}^{\sqrt{3}} \frac{\tan^{-1} x}{1+x^2} dx$$

- b) A tangent at the point of contact A of two circles (which touch externally) meets a common tangent, PQ to both circles, at R .



Prove that

- i) R is the midpoint of PQ
- ii) PQ subtends a right angle at A
- c) By using the expansions of $\cos(A+B)$ and $\cos(A-B)$
- i) find an expression for $\sin x \sin 3x$
- ii) hence evaluate $\int_0^{\frac{\pi}{4}} \sin x \sin 3x dx$

2**3****2****2****Question 4 (12 marks) - Start a new page**

- a) If the chance that any one of 6 telephone lines is busy at any instant is $\frac{1}{3}$
- i) What is the chance that exactly 4 of the lines are busy?
- ii) Determine the probability that at most two of the lines are busy
- b) The volume of a sphere is increasing at the rate of $5\text{cm}^3/\text{s}$.
At what rate is the surface area increasing when the radius is 20cm.
- c) A particle moves in a straight line and its position in metres at anytime t seconds is given by $x = 3 \cos 2t - 4 \sin 2t$
- i) by expressing the motion in terms of $A \cos(nt + \alpha)$
Show that the motion is simple harmonic.
- ii) Find the particle's greatest speed

2**2****3****3****2**

Question 5 (12 marks) - Start a new page**Marks**

- a) Use mathematical induction to prove the identity
 $1.2 + 2.3 + 3.4 + \dots + n(n+1) = \frac{1}{3} n(n+1)(n+2)$ **6**
hence determine the limit of $\frac{1}{n^3} \sum_{k=1}^n k(k+1)$ as n approaches infinity
- b) Newton's law of cooling states that the rate of change of the temperature θ of a body at any time t is proportional to the difference in temperature of the body and the temperature m of the surrounding medium, i.e. $\frac{d\theta}{dt} = k(\theta - m)$ where k is a constant.
- i) Show that $\theta = m + Ae^{kt}$ where A is a constant, satisfies this equation. **1**
- ii) If the temperature of the surrounding air is 40°C and the temperature of the body drops from 170°C to 105°C in 45 mins, find the temperature of the body in another 90 minutes. (to 2 decimal places) **3**
- iii) Find the time taken for the temperature of the body to drop to 80°C (to the nearest minute) **2**

Question 6 (12 marks) - Start a new page

- a) Solve $\log_e(\log_e x) = 0$ (in exact form) **1**
- b) Find the equation of the normal at the point $T(2at, at^2)$ on the parabola $x^2 = 4ay$. Hence determine the value(s) of t for the equations of the normals to this parabola to pass through the point $(-12a, 15a)$. **5**
- c) The acceleration of a particle P is given by the equation
 $\frac{d^2x}{dt^2} = 8x(x^2 + 1)$ where x is the displacement in cm, of P from a fixed point O after t seconds.
Initially, P is at the origin moving with velocity -2cm/s .
- i) Show that the speed of the particle is $2(x^2 + 1)\text{cm/s}$ and hence find an expression for x in terms of t **5**
- ii) Determine the displacement of P after $\frac{\pi}{8}$ seconds **1**

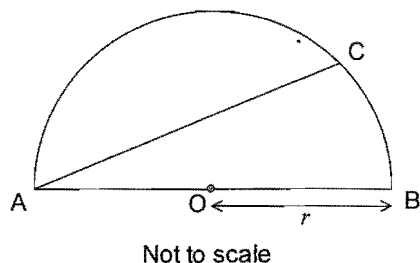
Question 7 (12 marks) - Start a new page

Marks

a) Evaluate $\int_0^1 \frac{e^{\cos^{-1} x}}{\sqrt{1-x^2}} dx$

2

b)



AB is the diameter of a semi circle with radius r

The chord AC divides the semicircle into two regions of equal area.

- i) By letting $\angle CAB = \theta$ radians
Prove that $2\theta + \sin 2\theta = \frac{\pi}{2}$

2

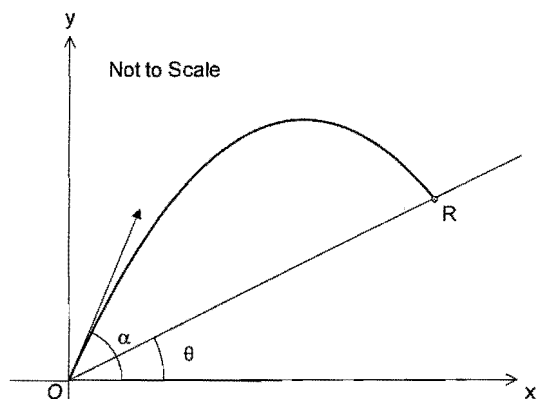
- ii) Show that $\theta = 0.4$ is a good approximation to the solution of $2\theta + \sin 2\theta = \frac{\pi}{2}$

1

- iii) Use Newton's method once to find an improved solution for the value of θ (to 2 significant figures)

2

c)



A stone is projected from O with velocity V at an angle α above the horizontal.

A straight road goes through O at an angle θ above the horizontal, where $\theta < \alpha$

The stone strikes the road at R .

Air resistance is to be ignored and the acceleration due to gravity is g

- i) Given that the equations of motion of the stone are

1

$$\left. \begin{aligned} x &= vt \cos \alpha \\ y &= vt \sin \alpha - \frac{gt^2}{2} \end{aligned} \right\} \text{ Do NOT prove these results.}$$

Show that the Cartesian equation for the motion is

$$y = x \tan \alpha - \frac{gx^2 \sec^2 \alpha}{2v^2}$$

- ii) If R is the point (x, y) express x and y in terms of RO and θ

2

Hence show that the range RO of the stone up the road is given by

$$RO = \frac{2v^2 \cos \alpha \sin(\alpha - \theta)}{g \cos^2 \theta}$$

- iii) Find an expression for α when RO is a maximum and interpret this result.

2

End of Paper

Question one.

$$\begin{aligned} \int_0^{\frac{\pi}{2}} \cos^2 x \, dx &= \frac{1}{2} \int_0^{\frac{\pi}{2}} (1 + \cos 2x) \, dx \quad \text{①} \\ &= \frac{1}{2} \left[x + \frac{\sin 2x}{2} \right]_0^{\frac{\pi}{2}} \\ &= \frac{1}{2} \left(\frac{\pi}{2} + \frac{\sin \pi}{2} - 0 \right) \\ &= \frac{\pi}{4} \quad \text{①} \end{aligned}$$

$$\begin{aligned} \text{b) } \cos 2\theta &= \sin \theta \quad 0 \leq \theta \leq 2\pi \\ 1 - 2\sin^2 \theta &= \sin \theta \\ 2\sin^2 \theta + \sin \theta - 1 &= 0 \quad \text{①} \\ (2\sin \theta + 1)(\sin \theta - 1) &= 0 \\ \sin \theta &= -\frac{1}{2} \text{ or } \sin \theta = 1 \quad \text{①} \\ \theta &= \frac{\pi}{6}, \frac{5\pi}{6} \quad \theta = \frac{3\pi}{2} \\ \therefore \theta &= \frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2} \quad \text{①} \end{aligned}$$

$$\begin{aligned} \text{c) } \frac{3x+4}{x-5} &\geq 2 \\ \frac{3x+4-2x+10}{x-5} &\geq 0 \quad \text{① some progression towards solution} \\ \frac{x+14}{x-5} &\geq 0 \quad (x-5) \geq 0 \quad (x-5)^2 \\ (x-5)(x+14) &\geq 0 \quad \leftarrow \begin{array}{c} \text{graph} \\ \text{number line} \end{array} \\ x &\leq -14, x \geq 5 \\ \text{but } x &\neq 5 \\ \therefore x &\leq -14, x > 5 \quad \text{①} \end{aligned}$$

$$\begin{aligned} \text{d) } \int_{-1}^4 \frac{w \, dw}{\sqrt{w+2}} & \quad \begin{array}{l} w = x^2 - 2 \quad dw = 2x \, dx \\ w = 14 \quad x^2 = 16 \quad x = 4 \\ w = -1 \quad x^2 = 1 \quad x = 1 \end{array} \\ \therefore \int_1^4 \frac{(x^2-2) \cdot 2x \, dx}{\sqrt{x^2-2}} &= 2 \int_1^4 \frac{(x^2-2) \, dx}{\sqrt{x^2-2}} \quad \text{①} \\ &= 2 \left[\frac{x^3}{3} - 2x \right]_1^4 \\ &= 2 \left(\frac{64}{3} - 8 - \left(\frac{1}{3} - 2 \right) \right) = \frac{30}{1} \end{aligned}$$

$$\begin{aligned} \text{e) } n &= 10 \\ \text{i) } {}^8C_4 &= 70 \quad \text{①} \\ \text{ii) } {}^9C_6 &= 84 \quad \text{①} \end{aligned}$$

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Question Two

$$\begin{aligned} \text{a) } & \text{Graph of } y = \frac{3x-5}{x-1} \quad \text{midpt } \left(\frac{-3+5}{2}, \frac{1+7}{2} \right) = (1, 4) \quad \text{①} \\ & \text{M}_{AE} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{7-4}{5-1} = \frac{3}{4} \\ & \tan \theta = \frac{|m_1 - m_2|}{1 + m_1 m_2} \\ & = \frac{\left| 2 - \frac{3}{4} \right|}{1 + 2 \times \frac{3}{4}} \quad \text{①} \quad \text{(ignore finding } \theta) \\ & = \frac{1}{2} \quad \text{①} \end{aligned}$$

$$\begin{aligned} \text{b) } 2x^3 + 5x - 3 &= 0 \\ \alpha + \beta + \gamma &= -\frac{b}{a} = \frac{5}{2} \\ \alpha\beta + \alpha\gamma + \beta\gamma &= \frac{c}{a} = -\frac{3}{2} \\ \alpha\beta\gamma &= -\frac{d}{a} = \frac{3}{2} \end{aligned}$$

$$\begin{aligned} \text{ii) } \alpha^2 + \beta^2 + \gamma^2 &= (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \alpha\gamma + \beta\gamma) \\ &= 0 - 2\left(-\frac{3}{2}\right) = 3 \quad \text{①} \\ \text{iii) } \frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} &= \frac{\beta\gamma + \alpha\gamma + \alpha\beta}{\alpha\beta\gamma} \\ &= \frac{5}{3} \quad \text{①} \end{aligned}$$

$$\begin{aligned} \text{c) } P(x) &= (x^2-1)(x-\alpha) \\ P(2) &= (4-1)(2-\alpha) = -9 \\ 6-3\alpha &= -9 \\ -3\alpha &= -15 \\ \alpha &= 5 \quad \text{①} \\ \therefore P(x) &= (x-1)(x+1)(x-5) \\ x &= 1, -1, 5. \quad \text{①} \end{aligned}$$

$$\begin{aligned} \text{Quest 2. Cont.} \\ \text{d) } \left(x^2 + \frac{4}{x}\right)^{30} \quad T_{k+1} &= \binom{n}{k} a^{n-k} b^k \\ &= \binom{30}{k} x^{60-2k} \cdot 4 \cdot x^{-k} \\ &= \binom{30}{k} 4 \cdot x^{60-3k} \end{aligned}$$

$$\begin{aligned} \text{i) independent of } x \text{ i.e. } x^0 \\ \therefore 0 &= 60-3k \quad \text{① statement to solve} \\ k &= 20 \\ \text{the term is } T_{21} \text{ or 21st term.} \end{aligned}$$

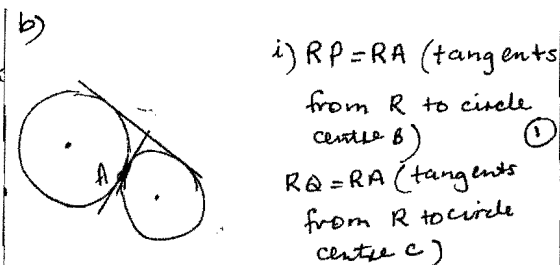
$$\begin{aligned} \text{ii) greatest co-eff.} \\ \frac{T_{k+1}}{T_k} &= \frac{n-k+1}{k} \cdot \frac{b}{a} \\ &= \frac{30-k+1}{k} \cdot \frac{4}{x^3} \quad \text{①} \\ \text{Coeff} &= \frac{30-k+1}{k} \cdot 4 > 1 \end{aligned}$$

$$\begin{aligned} 124-4k &> k \\ 124 &> 5k \\ k &< 24 \frac{4}{5} \\ \therefore k &= 24 \quad \text{①} \\ \text{the term is } T_{25} \text{ or 25th term.} \end{aligned}$$

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Question 3.

$$\begin{aligned} \text{a) } \frac{d}{dx} (\tan^{-1} x)^2 &= 2 \tan^{-1} x \times \frac{1}{1+x^2} \\ &= \frac{2 \tan^{-1} x}{1+x^2} \quad \text{①} \\ \int_{-1}^{\sqrt{3}} \frac{\tan^{-1} x}{1+x^2} \, dx &= \frac{1}{2} \int_{-1}^{\sqrt{3}} \frac{d}{dx} (\tan^{-1} x)^2 \, dx \\ &= \frac{1}{2} \left[(\tan^{-1} x)^2 \right]_{-1}^{\sqrt{3}} \quad \text{①} \\ &= \frac{1}{2} \left((\tan^{-1} \sqrt{3})^2 - (\tan^{-1} (-1))^2 \right) \\ &= \frac{1}{2} \left(\left(\frac{\pi}{3}\right)^2 - \left(-\frac{\pi}{4}\right)^2 \right) \\ &= \frac{7\pi^2}{288} \quad \text{①} \end{aligned}$$



$$\begin{aligned} \therefore RP &= RA \text{ (both equal to RA)} \\ \therefore R &\text{ is the midpoint of PA.} \end{aligned}$$

$$\begin{aligned} \text{ii) since } PR &= RA = RQ \text{ (from above)} \\ \therefore \text{ radii of circle centre } R \\ \text{then } PA &= PR + RA \text{ (a diameter)} \\ \therefore \angle PQA &= 90^\circ \text{ (L in a semi circle with centre R)} \quad \text{①} \end{aligned}$$

$$\begin{aligned} \text{--- or ---} \\ \text{in } \triangle APR \quad PR &= RA \text{ (above)} \\ \therefore \triangle APR &\text{ is isosceles (2 sides equal)} \\ \therefore \angle RPA &= \angle RAP \text{ (base Ls equal)} \quad \text{①} \\ &= x \end{aligned}$$

$$\begin{aligned} \text{in } \triangle ARQ \quad AR &= RQ \text{ (above)} \\ \therefore \triangle ARQ &\text{ is isos (2 sides equal)} \\ \therefore \angle RQA &= \angle QAR \text{ (base Ls equal)} \quad \text{①} \\ &= y \end{aligned}$$

$$\begin{aligned} \text{now in } \triangle PAQ \quad x+x+y+y &= 180 \text{ (L sum of } \triangle) \\ 2x+2y &= 180 \\ \therefore x+y &= 90^\circ \\ &= \angle PQA \quad \text{①} \end{aligned}$$

$$\begin{aligned} \text{c) } \cos(3x+x) &= \cos 3x \cos x - \sin 3x \sin x \\ \cos(3x-x) &= \cos 3x \cos x + \sin 3x \sin x \\ \therefore \cos(3x-x) - \cos(3x+x) &= 2 \sin 3x \sin x \quad \text{①} \\ \therefore \sin 3x \sin x &= \frac{1}{2} (\cos(2x) - \cos(4x)) \quad \text{①} \end{aligned}$$

Quest 3

c) cont

$$\begin{aligned} \text{ii) } \int_0^{\frac{\pi}{4}} \sin x \sin 3x dx &= \frac{1}{2} \int_0^{\frac{\pi}{4}} \cos 2x - \cos 4x dx \\ &= \frac{1}{2} \left[\frac{1}{2} \sin(2x) - \frac{1}{4} \sin(4x) \right]_0^{\frac{\pi}{4}} \\ &= \frac{1}{2} \left(\frac{1}{2} \sin \frac{\pi}{2} - \frac{1}{4} \sin \pi \right) \\ &\quad - \left(\frac{1}{2} \sin 0 - \frac{1}{4} \sin 0 \right) \\ &= \frac{1}{2} \left(\frac{1}{2} - 0 \right) \\ &= \frac{1}{4} \quad \text{①} \end{aligned}$$

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Question Four

a) $p = \frac{1}{3}$ $q = \frac{2}{3}$ $n = 6$
 ii) 4 lines busy $= {}^6C_4 \left(\frac{1}{3} \right)^4 \left(\frac{2}{3} \right)^2$ ①
 $= \frac{20}{243}$ ①

ii) $P(\text{at most 2 lines busy}) = P(0) + P(1) + P(2)$
 $= {}^6C_0 \left(\frac{2}{3} \right)^6 + {}^6C_1 \left(\frac{1}{3} \right)^1 \left(\frac{2}{3} \right)^5$
 $+ {}^6C_2 \left(\frac{1}{3} \right)^2 \left(\frac{2}{3} \right)^4$
 $= 0.6808 \dots$
 or $\frac{496}{729}$ ①

b) $V_{sp} = \frac{4}{3} \pi r^3$ $SA = 4\pi r^2$ $\frac{dV}{dt} = 5$
 $\frac{dV}{dr} = 4\pi r^2$ $\frac{dA}{dr} = 8\pi r$
 $\frac{dV}{dt} = \frac{dV}{dr} \cdot \frac{dr}{dt}$ $\frac{dA}{dr} = \frac{dA}{dr} \cdot \frac{dr}{dt}$
 $5 = 4\pi r^2 \cdot \frac{dr}{dt}$ $\frac{dA}{dr} = 8\pi r \times \frac{5}{4\pi r^2}$ ①
 $\frac{dr}{dt} = \frac{5}{4\pi r}$
 $\therefore SA \text{ is increasing at a rate of } 0.5 \text{ cm}^2/\text{s}$ ①

c) $x = 3 \cos 2t - 4 \sin 2t$

i) $x = A \cos(n\pi + \alpha)$ $n = 2$
 $A = \sqrt{16 + 9}$ $\tan \alpha = \frac{4}{3}$ (could be left $-\frac{4}{3}$)
 $= 5$ $\alpha = 0.927$ (radians)

$\therefore x = 5 \cos(2t + 0.927)$ ①
 $\dot{x} = -10 \sin(2t + 0.927)$
 $\ddot{x} = -20 \cos(2t + 0.927)$ ①
 $= -4x$ $\therefore$ in SHM with $n = 2$

ii) greatest speed $\Rightarrow \ddot{x} = 0$ or $x = 0$

$5 \cos(2t + 0.927) = 0$
 $\cos(2t + 0.927) = 0$
 $2t + 0.927 = \frac{\pi}{2}, \frac{3\pi}{2}, \dots$
 $2t = 0.643$
 $t = 0.3218 \dots$ ①

sub in to $\dot{x}$
 $\dot{x} = -10 \sin(2 \times 0.3218 + 0.927)$
 $= -10 \times 1$
 $= -10 \text{ m/sec}$ ①
 $\therefore \text{max speed} = |V| = 10 \text{ m/sec}$

Question Five

a) $1 \cdot 2 + 2 \cdot 3 + 3 \cdot 4 + \dots + n(n+1)$
 $S_n = \frac{1}{3} n(n+1)(n+2)$
 test $n = 1$
 $S_1 = LHS = 1(1+1)$ $RHS = \frac{1}{3}(2)(3)$
 $= 2$ $= \frac{1}{3} \times 6$
 $= 2$ ①
 $\therefore LHS = RHS$ true for $n = 1$

assume true for $n = k$
 ie $S_k = 1 \cdot 2 + 2 \cdot 3 + 3 \cdot 4 + \dots + k(k+1)$
 $= \frac{1}{3} k(k+1)(k+2)$

Prove true for $n = k+1$.

ie $S_{k+1} = S_k + T_{k+1}$

Quest 5 cont

a) cont.

now $T_{k+1} = (k+1)(k+2)$

$S_{k+1} = \frac{1}{3} (k+1)(k+2)(k+3)$

and

$S_k + T_{k+1} = \frac{1}{3} k(k+1)(k+2) + (k+1)(k+2)$ ①
 $= \frac{1}{3} k(k+1)(k+2) + \frac{3}{3} (k+1)(k+2)$
 $= \frac{(k+1)(k+2)}{3} (k+3)$
 $= \frac{1}{3} (k+1)(k+2)(k+3) = S_{k+1}$ ①

$\therefore$ assumed true for $n = k$ proved true for $n = k+1$ ①
 Since true for $n = 1$ now true by m.I. for $n = 1+1 = 2$, etc
 for all positive integers

now $\frac{1}{n^3} \sum_{k=1}^n k(k+1) = \frac{1}{n^3} (1(2) + 2(3) + 3(4) + \dots + n(n+1))$
 $= \frac{1}{n^3} \left(\frac{1}{3} n(n+1)(n+2) \right)$ ① from above

$\therefore \lim_{n \rightarrow \infty} \frac{1}{n^3} \sum_{k=1}^n k(k+1) = \lim_{n \rightarrow \infty} \frac{1}{3} \times \frac{1}{n^2} (n+1)(n+2)$
 $= \lim_{n \rightarrow \infty} \frac{1}{3} \times \frac{(n+1)}{n} \times \frac{(n+2)}{n}$
 $= \lim_{n \rightarrow \infty} \frac{1}{3} \left(1 + \frac{1}{n} \right) \left(1 + \frac{2}{n} \right)$
 $= \frac{1}{3} \times 1$ as $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$ and $\lim_{n \rightarrow \infty} \frac{2}{n} = 0$
 $= \frac{1}{3}$ ①

b) $\theta = m + A e^{kt}$

i) $\frac{d\theta}{dt} = k A e^{kt}$

but $A e^{kt} = \theta - m$ (from above)

$\therefore \frac{d\theta}{dt} = k(\theta - m)$ as required. ①

Ques 1b
sub for x into

$$x + ty = at^3 + 2at$$

at $x=1$

$$1 + y = a + 2a$$

$$x + y = a + 2a$$

$$x + y = 3a$$

at $x=-4$

$$x - 4y = a(-4)^3 + 2a(-4)$$

$$x - 4y = -72a$$

at $x=3$

$$x + 3y = 27a + 6a$$

$$x + 3y = 33a$$

taken out

c) $\frac{d^2x}{dt^2} = 8x(x^2+1)$ $x=0$ $v=-2\text{cm/s}$ $x=0$

since $\frac{d^2x}{dt^2} = \frac{d}{dx} \frac{1}{2} v^2$

$$\therefore \frac{d}{dx} \frac{1}{2} v^2 = 8x^3 + 8x$$

$$\frac{1}{2} v^2 = 2x^4 + 4x^2 + C$$

$x=0, v=-2$

$$\therefore \frac{1}{2} \times 4 = 0 + 0 + C$$

$$C = 2$$

$$\therefore \frac{1}{2} v^2 = 2x^4 + 4x^2 + 2$$

$$v^2 = 4x^4 + 8x^2 + 4$$

$$= 4(x^2 + 2x + 1)$$

$$= 4(x^2 + 1)^2$$

$$v = \pm 2(x^2 + 1)$$

$$\therefore \text{speed} = |v| = 2(x^2 + 1) \text{ cm/sec.}$$

now $\frac{dx}{dt} = \pm 2(x^2 + 1)$ ($\pm$ as nat speed)

$$\frac{dx}{dx} = \frac{\pm 1}{2(1+x^2)}$$

from initial conditions can assume $\frac{dx}{dt} = \frac{1}{2(1+x^2)}$

$$x = \pm \frac{1}{2} \tan^{-1} x + C$$

$x=0$ $x=0$

$$0 = \pm \frac{1}{2} \tan^{-1} 0 + C$$

$$C = 0$$

$$2t = \pm \tan^{-1} x$$

then $2t = -\tan^{-1} x$

$\therefore \tan(\pm 2t) = x$

ie $\pm \tan(2t) = x$

considering both cases.

$$x = \tan 2t$$

$$\dot{x} = 2 \sec^2 2t$$

at $t=0$ $\dot{x} = -2$

$$-2 = 2 \sec^2 0$$

$$-2 = 2 \times 1$$

$$-2 = -2$$

$\therefore$ not possible

$$\therefore x = -\tan 2t$$

not required

ii) at $t = \frac{\pi}{8}$

$$x = -\tan 2 \times \frac{\pi}{8}$$

$$= -\tan \frac{\pi}{4}$$

$$= -1$$

$\therefore$ P is 1cm to the left of the origin.

Question 7.

a) $\int_0^1 \frac{e^{\cos^{-1}x}}{\sqrt{1-x^2}} dx = [-e^{\cos^{-1}x}]_0^1$

$$= -e^{\cos^{-1}1} + e^{\cos^{-1}0}$$

$$= -e^0 + e^{\frac{\pi}{2}}$$

$$= e^{\frac{\pi}{2}} - 1$$

b) θ construct $\triangle AOC$ is isosceles (radii-sides equal)

$\angle COB = 2\theta$ (ext $\angle$ to $\triangle$)

$\angle AOC = \pi - 2\theta$

Area $\text{sector} = A_{\triangle} + A_{\text{sector}}$

i) $\frac{1}{2} \text{Semi circle} = \frac{1}{2} r^2 \sin(\pi - 2\theta) + \frac{1}{2} r^2 \times 2\theta$

$$\frac{\pi}{4} = \frac{1}{2} (\sin(\pi - 2\theta) + 2\theta)$$

$$\frac{\pi}{2} = \sin 2\theta + 2\theta$$

ii) $\theta = 0.4$

$$f(\theta) = 2\theta + \sin 2\theta - \frac{\pi}{2} = 0$$

$$f(0.4) = 0.8 + \sin 0.8 - \frac{\pi}{2} = -0.053$$

$$\therefore 0$$

Ques 5 Cont.

b) ii) $m = 40$ $\theta = 170$ $t=0$

$\theta = 105$ $t = 45$

$\therefore \theta = ?$ when $t = 90 + 45 = 135$

$$\theta = m + Ae^{kt}$$

at $t=0$

$$170 = 40 + Ae^0$$

$$A = 130$$

$$\theta = 40 + 130 e^{kt}$$

at $t=45$

$$105 = 40 + 130 e^{45k}$$

$$65 = 130 e^{45k}$$

$$\ln \frac{65}{130} = 45k \ln e$$

$$k = (\ln \frac{65}{130}) \div 45$$

$$\approx -0.0154 \dots$$

at $t=135$

$$\theta = 40 + 130 e^{135k}$$

$$= 56.25^\circ \text{C}$$

iii) $t = ?$

$$80 = 40 + 130 e^{kt}$$

$$40 = 130 e^{kt}$$

$$\ln \frac{4}{13} = kt \ln e$$

$$t = \ln(\frac{4}{13}) \div k$$

$$= 76.53$$

$$= 77 \text{ min.}$$

Question 6

a) let $M = \log_e x$

$$\therefore \log_e M = 0$$

$$e^0 = M$$

$$M = 1$$

$$\therefore \log_e x = 1$$

$$e^1 = x$$

b) $x^2 = 4ay$ $T(2at, at^2)$

$$y = \frac{x^2}{4a}$$

$$\frac{dy}{dx} = \frac{2x}{4a}$$

at $x=2at$ $m = t$

For the normal $m_2 = -\frac{1}{t}$

equation: $y - y_1 = m(x - x_1)$

$$y - at^2 = -\frac{1}{t}(x - 2at)$$

$$ty - at^3 = -x + 2at$$

$$\therefore x + ty = at^3 + 2at$$

passes thru $(-12a, 15a)$

$$\therefore -12a + 15at = at^3 + 2at$$

$$at^3 - 13at + 12a = 0$$

$$t^3 - 13t + 12 = 0$$

Find factor: try 1, -1, etc.

$$P(t) = t^3 - 13t + 12$$

$$P(1) = 1 - 13 + 12 = 0 \therefore (t-1)$$

is a factor

$$\begin{array}{r} t^3 + t^2 - 12 \\ t-1 \overline{) t^3 - 13t + 12} \\ \underline{t^3 - t^2} \\ t^2 - 13t + 12 \\ \underline{t^2 - t} \\ -12t + 12 \\ \underline{-12t + 12} \\ 0 \end{array}$$

$$\therefore (t-1)(t+4)(t-3) = 0$$

$$\therefore t = 1, -4, 3$$

Quest 7. cont

b) (ii)

$$f(\theta) = \sin 2\theta + 2\theta - \frac{\pi}{2}$$

$$f'(\theta) = 2\cos 2\theta + 2$$

$$f'(0.4) = 2\cos 0.8 + 2$$

$$\therefore \theta_2 = \theta_1 - \frac{f(\theta_1)}{f'(\theta_1)} \quad \text{①}$$

$$= 0.4 - \frac{\sin 0.8 + 0.8 - \frac{\pi}{2}}{2\cos 0.8 + 2}$$

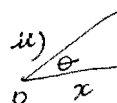
$$= 0.4157 \dots \text{any approx} \quad \text{①}$$

c) $x = vt \cos \alpha$ $y = vt \sin \alpha - \frac{gt^2}{2}$

d) $t = \frac{x}{v \cos \alpha} \rightarrow y = v \left(\frac{x}{v \cos \alpha} \right) \sin \alpha - g \left(\frac{x}{v \cos \alpha} \right)^2$

$$\text{①} = x \tan \alpha - \frac{g x^2}{2 v^2 \cos^2 \alpha}$$

$$R \quad y = x \tan \alpha - \frac{g x^2 \sec^2 \alpha}{2 v^2}$$



$$\cos \theta = \frac{x}{R}$$

$$x = R \cos \theta$$

$$\sin \theta = \frac{y}{R}$$

$$y = R \sin \theta$$

① either

Test max

$$\frac{d^2 RO}{d\alpha^2} = \frac{2v^2 x - 2 \sin(2\alpha - \theta)}{g \cos^2 \theta}$$

$$= \frac{-4v^2 \sin(2\alpha - \theta)}{g \cos^2 \theta}$$

$$\text{at } \alpha = \frac{1}{2}(\theta + \frac{\pi}{2})$$

$$\frac{d^2 RO}{d\alpha^2} = \frac{-4v^2 \sin(2 \times \frac{1}{2}(\theta + \frac{\pi}{2}) - \theta)}{g \cos^2 \theta}$$

$$= \frac{-4v^2 \times 1}{g \cos^2 \theta} < 0 \therefore \text{max.}$$

Since RO is a max for $\alpha = \frac{1}{2}(\theta + \frac{\pi}{2})$

① α bisects the angle $(\theta + \frac{\pi}{2})$
the end! 12

sub in to cartesian equation

$$R \sin \theta = R \cos \theta \tan \alpha - \frac{g(R \cos \theta)^2 \sec^2 \alpha}{2 v^2}$$

$$R \sin \theta = \frac{R \cos \theta \sin \alpha}{\cos \alpha} - \frac{g \cdot R^2 \cos^2 \theta}{2 v^2 \cos^2 \alpha}$$

$$R \sin \theta \cdot \cos^2 \alpha = R \cos \theta \sin \alpha \cos \alpha - \frac{g \cdot R^2 \cos^2 \theta}{2 v^2}$$

$$\div R \quad \sin \theta \cdot \cos^2 \alpha = \cos \theta \sin \alpha \cos \alpha - \frac{g \cdot R \cos^2 \theta}{2 v^2}$$

$$\frac{g \cos^2 \theta}{2 v^2} \cdot R = \cos \theta \sin \alpha \cos \alpha - \sin \theta \cos^2 \alpha \quad \text{①}$$

$$= \cos \alpha (\sin \alpha \cos \theta - \cos \alpha \sin \theta)$$

$$= \cos \alpha \cdot \sin(\alpha - \theta)$$

$$R = \frac{2 v^2 \cos \alpha \sin(\alpha - \theta)}{g \cos^2 \theta}$$